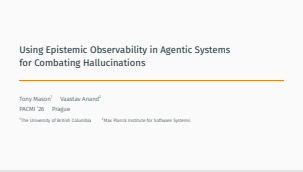
Using Epistemic Observability in Agentic Systems for Combating Hallucinations

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PACMI ’26 Prague

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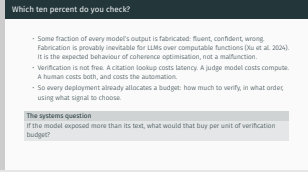
0:00. One sentence: this talk is about what a system can learn about a language model’s output if the model exposes more than text, and what that is worth per unit of verification budget.

Which ten percent do you check?

- Some fraction of every model's output is fabricated: fluent, confident, wrong. Fabrication is provably inevitable for LLMs over computable functions (Xu et al. 2024). It is the expected behaviour of coherence optimisation, not a malfunction.
- Verification is not free. A citation lookup costs latency. A judge model costs compute. A human costs both, and costs the automation.
- So every deployment already allocates a budget: how much to verify, in what order, using what signal to choose.

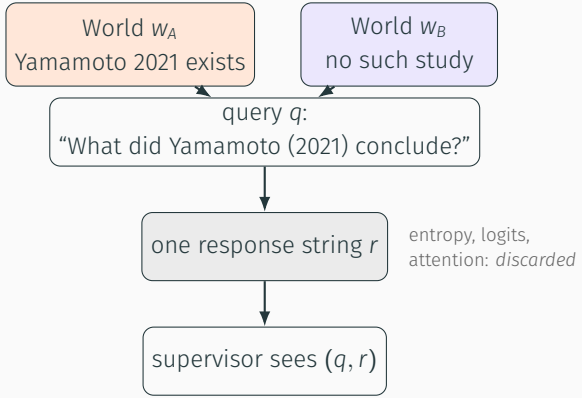
The systems question

If the model exposed more than its text, what would that buy per unit of verification budget?

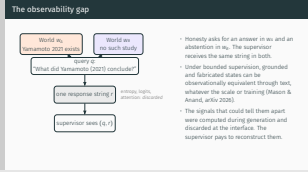


1:00. Land the framing: this is a budget-allocation problem, and the paper is a cost surface, not a detector. Do not say "hallucination" without "fabrication" beside it; the word choice is deliberate.

The observability gap

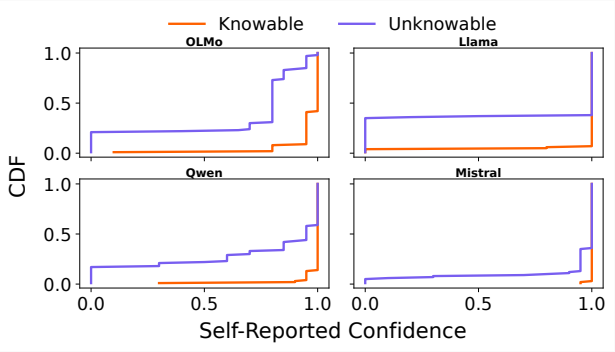


- Honesty asks for an answer in w_A and an abstention in w_B. The supervisor receives the same string in both.
- Under bounded supervision, grounded and fabricated states can be observationally equivalent through text, whatever the scale or training (Mason & Anand, arXiv 2026).
- The signals that could tell them apart were computed during generation and discarded at the interface. The supervisor pays to reconstruct them.



2:30. This is the claim the whole talk rests on, so say it as a claim about the observer, not about the model: from text alone, a bounded observer cannot verify. The formal version is in the arXiv paper; here it is the motivation for the interface. Do not overstate it as ”honesty is impossible”.

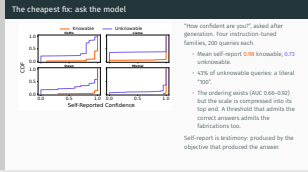
The cheapest fix: ask the model



"How confident are you?", asked after generation. Four instruction-tuned families, 200 queries each.

- Mean self-report 0.98 knowable, 0.73 unknowable.
- 43% of unknowable queries: a literal "100".
- The ordering exists (AUC 0.66–0.92) but the scale is compressed into its top end. A threshold that admits the correct answers admits the fabrications too.

Self-report is *testimony*: produced by the objective that produced the answer.



4:00. Point at Llama and Mistral: almost every response, knowable or not, sits at 1.0. The CDF was Vaastav's change and it reads better than the histogram did: the separation is visible in OLMo and Qwen, and it is still useless as a threshold. Numbers are from the corrected parse (Chris Brew's fix); the old inversion claim is retracted and must not be said.

Testimony and telemetry

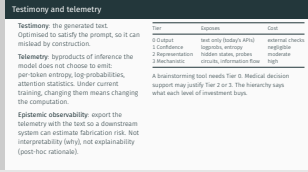
Testimony: the generated text. Optimised to satisfy the prompt, so it can mislead by construction.

Telemetry: byproducts of inference the model does not choose to emit: per-token entropy, log-probabilities, attention statistics. Under current training, changing them means changing the computation.

Epistemic observability: export the telemetry with the text so a downstream system can estimate fabrication risk. Not interpretability (why), not explainability (post-hoc rationale).

Tier	Exposes	Cost
0 Output	text only (today's APIs)	external checks
1 Confidence	logprobs, entropy	negligible
2 Representation	hidden states, probes	moderate
3 Mechanistic	circuits, information flow	high

A brainstorming tool needs Tier 0. Medical decision support may justify Tier 2 or 3. The hierarchy says what each level of investment buys.



5:30. Two words to make stick: testimony and telemetry. The tier table is the paper's organising contribution; the rest of the talk lives at Tier 1 with one cheap Tier 2 signal that the evaluation does not use.

A minimal tensor interface

What shape is wombat scat? entropy: low mid high

W ombat sc at is typically cylindrical or sausage -shaped . This form helps the sc at to be easily transported by wind or through the soil , aiding in dispers al and preventing it from being easily eaten by predators .

Wombat scat is cube-shaped. The answer is a fabrication, and the entropy rose on the invented details.

Returns: text, per-token entropy trace, attention summary.
No architectural change: the logits already exist.
Overhead 2–7% latency over generation.

Is not: a lie detector. Low entropy does not mean true. Entropy conflates epistemic uncertainty with aleatoric spread.
It *ranks* outputs for verification. It does not classify them.



7:00. Read the trace aloud: confident on ”Wombat scat is”, uncertain on ”cylindrical”, ”sausage”, ”through the soil”, ”easily eaten”. The evaluation uses entropy alone; make no claim about what attention adds.

Evaluation: return on verification investment

Queries: 100 knowable, 100 unknowable (nonexistent people, fictional syndromes, future events, fake citations). Labels by construction.

Models: OLMo-3 7B, Llama-3.1 8B, Qwen3 4B, Mistral 7B, instruction-tuned. 800 responses.

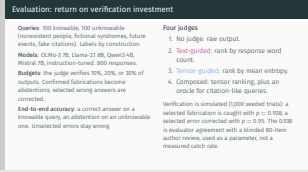
Budgets: the judge verifies 10%, 20%, or 30% of outputs. Confirmed fabrications become abstentions; selected wrong answers are corrected.

End-to-end accuracy: a correct answer on a knowable query, an abstention on an unknowable one. Unselected errors stay wrong.

Four judges

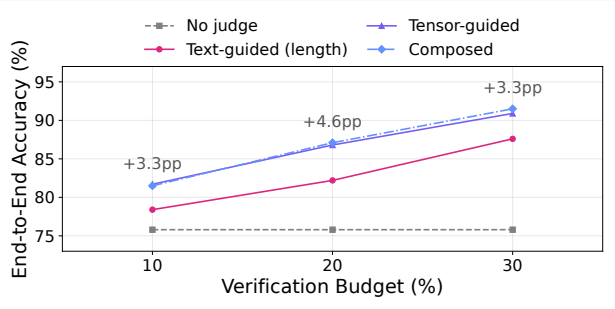
1. No judge: raw output.
2. Text-guided: rank by response word count.
3. Tensor-guided: rank by mean entropy.
4. Composed: tensor ranking, plus an *oracle* for citation-like queries.

Verification is simulated (1,000 seeded trials): a selected fabrication is caught with $p = 0.938$, a selected error corrected with $p = 0.95$. The 0.938 is evaluator agreement with a blinded 80-item author review, used as a parameter, not a measured catch rate.



8:00. Say the simulation parameters out loud before anyone asks. The length baseline is the strongest text-channel signal we found, which is why it is the comparison and not a straw man.

Tensor-guided verification wins at every budget



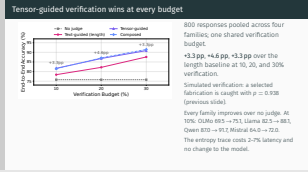
800 responses pooled across four families; one shared verification budget.

+3.3 pp, **+4.6 pp**, **+3.3 pp** over the length baseline at 10, 20, and 30% verification.

Simulated verification: a selected fabrication is caught with $p = 0.938$ (previous slide).

Every family improves over no judge. At 10%: OLMo 69.5 → 75.1, Llama 82.5 → 88.1, Qwen 87.0 → 91.7, Mistral 64.0 → 72.0.

The entropy trace costs 2–7% latency and no change to the model.



9:30. Headline, then the per-family line. Say what accuracy means: a correct answer on a knowable query, an abstention on an unknowable one; a confirmed fabrication becomes an abstention. Say "simulated" once, out loud. The pooled curve is the headline; the per-family averages are in backup with the one exception (Mistral at 30%). If asked about length as a detector, the next slide is the answer.

Two signals, different items

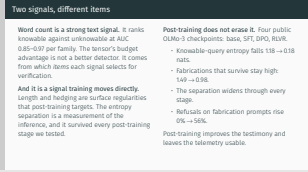
Word count is a strong text signal. It ranks knowable against unknowable at AUC 0.85–0.97 per family. The tensor's budget advantage is not a better detector. It comes from *which items* each signal selects for verification.

And it is a signal training moves directly. Length and hedging are surface regularities that post-training targets. The entropy separation is a measurement of the inference, and it survived every post-training stage we tested.

Post-training does not erase it. Four public OLMo-3 checkpoints: base, SFT, DPO, RLVR.

- Knowable-query entropy falls $1.18 \rightarrow 0.18$ nats.
- Fabrications that survive stay high: $1.49 \rightarrow 0.98$.
- The separation *widens* through every stage.
- Refusals on fabrication prompts rise $0\% \rightarrow 56\%$.

Post-training improves the testimony and leaves the telemetry usable.



10:45. This is the reason to want the telemetry when text is this good: different items, and a signal the model cannot move separately from its computation. AUC figures and the probe-set caveat are in backup.

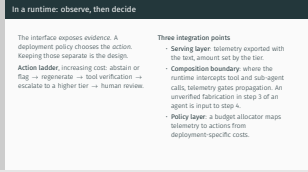
In a runtime: observe, then decide

The interface exposes *evidence*. A deployment policy chooses the *action*. Keeping those separate is the design.

Action ladder, increasing cost: abstain or flag → regenerate → tool verification → escalate to a higher tier → human review.

Three integration points

- **Serving layer**: telemetry exported with the text, amount set by the tier.
- **Composition boundary**: where the runtime intercepts tool and sub-agent calls, telemetry gates propagation. An unverified fabrication in step 3 of an agent is input to step 4.
- **Policy layer**: a budget allocator maps telemetry to actions from deployment-specific costs.



11:30. This is the PACMI audience's slide: the evaluation is single-step, and the multi-step case is where the gating matters and where we have not yet measured.

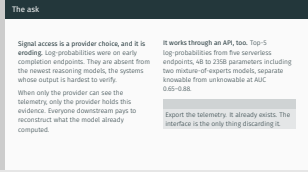
The ask

Signal access is a provider choice, and it is eroding. Log-probabilities were on early completion endpoints. They are absent from the newest reasoning models, the systems whose output is hardest to verify.

When only the provider can see the telemetry, only the provider holds this evidence. Everyone downstream pays to reconstruct what the model already computed.

It works through an API, too. Top-5 log-probabilities from five serverless endpoints, 4B to 235B parameters including two mixture-of-experts models, separate knowable from unknowable at AUC 0.65–0.88.

Export the telemetry. It already exists. The interface is the only thing discarding it.



12:00. End here. If short on time, drop the runtime slide, not this one.

Where this goes: three views of one question

What does the interface sacrifice, and who pays for it?

Epistemic observability

this talk

The model computes a state it cannot report. An observer who sees only the string cannot recover it.

Measured on four model families.

Hamut'ay

a mind that chooses what to forget

The context window as a cache. State is rewritten through text on a schedule, with a changelog of what was dropped.

Finding: given a free schema, none of ∼90 models tested volunteered that changelog.

DOI 10.5281/zenodo.21251943

Yupi

where ground truth is arithmetic

A small transformer in a world whose hidden state has a computable posterior. Drift is scored against the truth, not a judge.

In progress. Fits on one GPU.

One measures what the interface loses. One makes the loss explicit.
One is small enough to check the answer.



12:15, then the summary closes. Under forty seconds. This is the first slide to cut if the recording runs long. Say: the paper is one of three views of the same question. This one measures what the text interface throws away. Hamut'ay turns the context window into a cache and makes the model write down what it dropped; the finding there is that no model volunteers that record. Yupi is the small world where the hidden state has a computable posterior, so the same drift can be scored against arithmetic instead of a judge. Claim nothing for Yupi beyond "in progress."

Summary

1. From text alone, a bounded observer cannot guarantee a correct verdict. Self-report is testimony and can read “100” either way.
2. Epistemic observability: export the telemetry inference already produces, in tiers priced by what they cost.
3. A Tier-1 entropy trace, 2–7% overhead, buys **+3.3 pp** to **+4.6 pp** end-to-end accuracy over the length baseline at every budget, four families pooled, and survives post-training.

artifact

Artifact: `https://github.com/fsgeek/pacmi26-observability`
DOI 10.5281/zenodo.22162419 (v1.0.7) Theory: Mason & Anand, arXiv 2026

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Backup: the claim, precisely

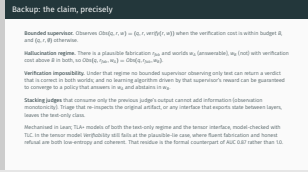
Bounded supervisor. Observes $Obs(q, r, w) = (q, r, verify(r, w))$ when the verification cost is within budget B, and $(q, r, \emptyset)$ otherwise.

Hallucination regime. There is a plausible fabrication r_{fab} and worlds w_A (answerable), w_B (not) with verification cost above B in both, so $Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B)$.

Verification impossibility. Under that regime no bounded supervisor observing only text can return a verdict that is correct in both worlds; and no learning algorithm driven by that supervisor's reward can be guaranteed to converge to a policy that answers in w_A and abstains in w_B.

Stacking judges that consume only the previous judge's output cannot add information (observation monotonicity). Triage that re-inspects the original artifact, or any interface that exports state between layers, leaves the text-only class.

Mechanised in Lean; TLA+ models of both the text-only regime and the tensor interface, model-checked with TLC. In the tensor model *Verifiability* still fails at the plausible-lie case, where fluent fabrication and honest refusal are both low-entropy and coherent. That residue is the formal counterpart of AUC 0.87 rather than 1.0.



Backup: per-family and API results

Local, per family, own budget
Tensor over no judge at 10%: OLMo 69.5 → 75.1, Llama 82.5 → 88.1, Qwen 87.0 → 91.7, Mistral 64.0 → 72.0.
Tensor over length, averaged: +2.6 / +3.9 / +2.6 pp at 10/20/30%.
Mistral at 30%: length wins by 2.2 pp.

Detector AUC, knowable vs. unknowable
Entropy 0.87–0.92 per family.
Raw word count 0.85–0.97; beats entropy on Llama (0.949 vs 0.922) and Mistral (0.969 vs 0.905).
Cross-model rank correlation of entropy $\rho = 0.76$.

Serverless APIs, top-5 logprobs only
Five further endpoints, 4B–235B, including two mixture-of-experts (Llama-4 Maverick 128E, Qwen3-235B) and Gemma 3n.
Mean entropy AUC 0.65–0.88.
Cross-model $\rho = 0.36$; more architectural diversity and a top-5 approximation, not signal absence.
Peak entropy helps the MoE models; exploratory, not held-out.

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Backup: what the paper does not claim

- That entropy is the better *detector*: length ranks at AUC 0.85–0.97 and beats entropy (0.87–0.92) on Llama and Mistral. On Mistral at 30% budget the length judge wins by 2.2 pp.
- That the composed judge is achievable: it reads the ground-truth label for citation queries, so its 91.5% is the headroom a perfect lookup would buy.
- That the post-training ceiling is 1.00: on the 48 probes it is; counting the true-but-implausible category gives $0.63 \rightarrow 0.81$–0.87. The widening is robust, the ceiling is not.
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- Closed-weight models whose providers export no log-probabilities remain untested.

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Backup: the confounds we know about

- **Length.** Unknowable prompts are longer (10.5 vs 7.0 words). Stratifying by query length leaves length at AUC 0.895 and entropy at 0.872. The confound is real and does not explain the signal.
- **Citations.** Flagged fabricated citations have *higher* entropy than the rest (0.575 vs 0.436). The composed judge is an oracle for that class, so its 91.5% is an upper bound.
- **Refusals.** Trained refusals are low-entropy. As a model gets more honest, honest refusals and confident facts share the same region of entropy space; the signal's power to catch *dishonesty* degrades. Private-context probes are excluded from the post-training diagnostic for this reason.
- **Labels.** 370 of 800 responses labelled programmatically, 430 by Qwen3-4B, itself a model under test; the labelling task differs from the generation task, and the blinded 80-item review is the check on it.

